

1000 AI Projects — Foundation to GenAI & Agentic AI

LLM · UI (React / Streamlit) · Backend (Python FastAPI) · AI Frameworks (LangGraph / LangChain / AutoGen / CrewAI / LlamaIndex / DSPy) · Vector DB & Infra

Master in AI Engineering — Build production-grade GenAI, Agentic AI, RAG & MLOps systems.
From LLM foundations to multi-agent orchestration — the complete hands-on path for AI engineers.

Created By Balawant M Kadam
Exported: 20/9/2026, 9:04:28 pm
Projects in export: 120 of 1000
Filters: Level: foundation | Type: All | Domain: All | Framework: All

#0001

TRADITIONAL

FOUNDATION

Clinical Notes Explorer

Healthcare

Build a Explorer using scikit-learn that applies classical machine learning and lightweight LLM calls to clinical notes within the Healthcare domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [scikit-learn](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA FAISS · Neo4j

#0002

TRADITIONAL

FOUNDATION

Diagnostic Imaging Classifier

Healthcare

Build a Classifier using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to diagnostic imaging within the Healthcare domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [FastAPI](#) [LightGBM](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0003

TRADITIONAL

FOUNDATION

Patient Triage Predictor

Healthcare

Build a Predictor using pandas that applies classical machine learning and lightweight LLM calls to patient triage within the Healthcare domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA Pinecone · DuckDB

#0004

TRADITIONAL

FOUNDATION

Drug Interaction Checker Dashboard

Healthcare

Build a Dashboard using NumPy that applies classical machine learning and lightweight LLM calls to drug interaction checker within the Healthcare domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [Docker](#) [DuckDB](#)

VECTOR / INFRA Weaviate · LanceDB

#0005

TRADITIONAL

FOUNDATION

EHR Records Basic Chatbot

Healthcare

Build a Basic Chatbot using scikit-learn that applies classical machine learning and lightweight LLM calls to ehr records within the Healthcare domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [Gradio](#) [MLflow](#)

VECTOR / INFRA Qdrant · OpenSearch

#0006

TRADITIONAL

FOUNDATION

Remote Patient Monitoring Rule Engine

Healthcare

Build a Rule Engine using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to remote patient monitoring within the Healthcare domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS

Hugging Face Transformers

scikit-learn

XGBoost

Streamlit

VECTOR / INFRA

Milvus · FAISS

#0007

TRADITIONAL

FOUNDATION

Fraud Detection Data Pipeline

Finance & Banking

Build a Data Pipeline using pandas that applies classical machine learning and lightweight LLM calls to fraud detection within the Finance & Banking domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS

pandas

LightGBM

FastAPI

VECTOR / INFRA

pgvector · ChromaDB

#0008

TRADITIONAL

FOUNDATION

Credit Scoring Visualizer

Finance & Banking

Build a Visualizer using NumPy that applies classical machine learning and lightweight LLM calls to credit scoring within the Finance & Banking domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS

NumPy

pandas

Polars

Pydantic

VECTOR / INFRA

Neo4j · Pinecone

#0009

TRADITIONAL

FOUNDATION

Portfolio Analytics Checker

Finance & Banking

Build a Checker using scikit-learn that applies classical machine learning and lightweight LLM calls to portfolio analytics within the Finance & Banking domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [scikit-learn](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0010

TRADITIONAL

FOUNDATION

Loan Underwriting Tutorial Notebook

Finance & Banking

Build a Tutorial Notebook using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to loan underwriting within the Finance & Banking domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [MLflow](#) [Gradio](#)

VECTOR / INFRA DuckDB · Qdrant

#0011

TRADITIONAL

FOUNDATION

Transaction Reconciliation Explorer

Finance & Banking

Build a Explorer using pandas that applies classical machine learning and lightweight LLM calls to transaction reconciliation within the Finance & Banking domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [pandas](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA LanceDB · Milvus

#0012

TRADITIONAL

FOUNDATION

Market Sentiment Classifier

Finance & Banking

Build a Classifier using NumPy that applies classical machine learning and lightweight LLM calls to market sentiment within the Finance & Banking domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	NumPy pandas FastAPI LightGBM
VECTOR / INFRA	OpenSearch · pgvector

#0013

TRADITIONAL

FOUNDATION

Product Recommendations Predictor

Retail & E-commerce

Build a Predictor using scikit-learn that applies classical machine learning and lightweight LLM calls to product recommendations within the Retail & E-commerce domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	scikit-learn Pydantic Polars
VECTOR / INFRA	FAISS · Neo4j

#0014

TRADITIONAL

FOUNDATION

Demand Forecasting Dashboard

Retail & E-commerce

Build a Dashboard using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to demand forecasting within the Retail & E-commerce domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	Hugging Face Transformers scikit-learn Docker DuckDB
VECTOR / INFRA	ChromaDB · Elasticsearch

#0015

TRADITIONAL

FOUNDATION

Visual Search Basic Chatbot

Retail & E-commerce

Build a Basic Chatbot using pandas that applies classical machine learning and lightweight LLM calls to visual search within the Retail & E-commerce domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [Gradio](#) [MLflow](#)

VECTOR / INFRA Pinecone · DuckDB

#0016

TRADITIONAL

FOUNDATION

Dynamic Pricing Rule Engine

Retail & E-commerce

Build a Rule Engine using NumPy that applies classical machine learning and lightweight LLM calls to dynamic pricing within the Retail & E-commerce domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [XGBoost](#) [Streamlit](#)

VECTOR / INFRA Weaviate · LanceDB

#0017

TRADITIONAL

FOUNDATION

Cart Abandonment Data Pipeline

Retail & E-commerce

Build a Data Pipeline using scikit-learn that applies classical machine learning and lightweight LLM calls to cart abandonment within the Retail & E-commerce domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [LightGBM](#) [FastAPI](#)

VECTOR / INFRA Qdrant · OpenSearch

#0018

TRADITIONAL

FOUNDATION

Inventory Optimization Visualizer

Retail & E-commerce

Build a Visualizer using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to inventory optimization within the Retail & E-commerce domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [Polars](#) [Pydantic](#)

VECTOR / INFRA Milvus · FAISS

#0019

TRADITIONAL

FOUNDATION

Adaptive Learning Paths Checker

Education

Build a Checker using pandas that applies classical machine learning and lightweight LLM calls to adaptive learning paths within the Education domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [pandas](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA pgvector · ChromaDB

#0020

TRADITIONAL

FOUNDATION

Essay Grading Tutorial Notebook

Education

Build a Tutorial Notebook using NumPy that applies classical machine learning and lightweight LLM calls to essay grading within the Education domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS [NumPy](#) [pandas](#) [MLflow](#) [Gradio](#)

VECTOR / INFRA Neo4j · Pinecone

#0021

TRADITIONAL

FOUNDATION

Tutoring Assistant Explorer

Education

Build a Explorer using scikit-learn that applies classical machine learning and lightweight LLM calls to tutoring assistant within the Education domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [scikit-learn](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0022

TRADITIONAL

FOUNDATION

Curriculum Generator Classifier

Education

Build a Classifier using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to curriculum generator within the Education domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [FastAPI](#) [LightGBM](#)

VECTOR / INFRA DuckDB · Qdrant

#0023

TRADITIONAL

FOUNDATION

Student Engagement Analytics Predictor

Education

Build a Predictor using pandas that applies classical machine learning and lightweight LLM calls to student engagement analytics within the Education domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [pandas](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA LanceDB · Milvus

#0024

TRADITIONAL

FOUNDATION

Language Learning Dashboard

Education

Build a Dashboard using NumPy that applies classical machine learning and lightweight LLM calls to language learning within the Education domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS [NumPy](#) [pandas](#) [Docker](#) [DuckDB](#)

VECTOR / INFRA OpenSearch · pgvector

#0025

TRADITIONAL

FOUNDATION

Predictive Maintenance Basic Chatbot

Manufacturing

Build a Basic Chatbot using scikit-learn that applies classical machine learning and lightweight LLM calls to predictive maintenance within the Manufacturing domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [scikit-learn](#) [Gradio](#) [MLflow](#)

VECTOR / INFRA FAISS · Neo4j

#0026

TRADITIONAL

FOUNDATION

Quality Inspection Rule Engine

Manufacturing

Build a Rule Engine using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to quality inspection within the Manufacturing domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [XGBoost](#) [Streamlit](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0027

TRADITIONAL

FOUNDATION

Supply Chain Planning Data Pipeline

Manufacturing

Build a Data Pipeline using pandas that applies classical machine learning and lightweight LLM calls to supply chain planning within the Manufacturing domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Gemini 2.0 Flash
UI	Gradio
BACKEND	Python · Flask + Gunicorn
AI FRAMEWORKS	pandas LightGBM FastAPI
VECTOR / INFRA	Pinecone · DuckDB

#0028

TRADITIONAL

FOUNDATION

Robotic Process Control Visualizer

Manufacturing

Build a Visualizer using NumPy that applies classical machine learning and lightweight LLM calls to robotic process control within the Manufacturing domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Mistral 7B
UI	Streamlit + Pandas
BACKEND	Python · FastAPI + Pandas
AI FRAMEWORKS	NumPy pandas Polars Pydantic
VECTOR / INFRA	Weaviate · LanceDB

#0029

TRADITIONAL

FOUNDATION

Defect Detection Checker

Manufacturing

Build a Checker using scikit-learn that applies classical machine learning and lightweight LLM calls to defect detection within the Manufacturing domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Phi-4 mini
UI	Jupyter + ipywidgets
BACKEND	Python · Jupyter + FastAPI
AI FRAMEWORKS	scikit-learn DuckDB Docker
VECTOR / INFRA	Qdrant · OpenSearch

#0030

TRADITIONAL

FOUNDATION

Factory Digital Twin Tutorial Notebook

Manufacturing

Build a Tutorial Notebook using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to factory digital twin within the Manufacturing domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	Hugging Face Transformers scikit-learn MLflow Gradio
VECTOR / INFRA	Milvus · FAISS

#0031

TRADITIONAL

FOUNDATION

Route Optimization Explorer

Logistics & Supply Chain

Build a Explorer using pandas that applies classical machine learning and lightweight LLM calls to route optimization within the Logistics & Supply Chain domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	pandas Streamlit XGBoost
VECTOR / INFRA	pgvector · ChromaDB

#0032

TRADITIONAL

FOUNDATION

Fleet Management Classifier

Logistics & Supply Chain

Build a Classifier using NumPy that applies classical machine learning and lightweight LLM calls to fleet management within the Logistics & Supply Chain domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	NumPy pandas FastAPI LightGBM
VECTOR / INFRA	Neo4j · Pinecone

#0033

TRADITIONAL

FOUNDATION

Warehouse Robotics Predictor

Logistics & Supply Chain

Build a Predictor using scikit-learn that applies classical machine learning and lightweight LLM calls to warehouse robotics within the Logistics & Supply Chain domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [scikit-learn](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0034

TRADITIONAL

FOUNDATION

Last-Mile Delivery Dashboard

Logistics & Supply Chain

Build a Dashboard using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to last-mile delivery within the Logistics & Supply Chain domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [Docker](#) [DuckDB](#)

VECTOR / INFRA DuckDB · Qdrant

#0035

TRADITIONAL

FOUNDATION

Freight Document Processing Basic Chatbot

Logistics & Supply Chain

Build a Basic Chatbot using pandas that applies classical machine learning and lightweight LLM calls to freight document processing within the Logistics & Supply Chain domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [pandas](#) [Gradio](#) [MLflow](#)

VECTOR / INFRA LanceDB · Milvus

#0036

TRADITIONAL

FOUNDATION

Inventory Replenishment Rule Engine

Logistics & Supply Chain

Build a Rule Engine using NumPy that applies classical machine learning and lightweight LLM calls to inventory replenishment within the Logistics & Supply Chain domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS [NumPy](#) [pandas](#) [XGBoost](#) [Streamlit](#)

VECTOR / INFRA OpenSearch · pgvector

#0037

TRADITIONAL

FOUNDATION

Threat Detection Data Pipeline

Cybersecurity

Build a Data Pipeline using scikit-learn that applies classical machine learning and lightweight LLM calls to threat detection within the Cybersecurity domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [scikit-learn](#) [LightGBM](#) [FastAPI](#)

VECTOR / INFRA FAISS · Neo4j

#0038

TRADITIONAL

FOUNDATION

Phishing Analysis Visualizer

Cybersecurity

Build a Visualizer using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to phishing analysis within the Cybersecurity domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [Polars](#) [Pydantic](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0039

TRADITIONAL

FOUNDATION

Vulnerability Triage Checker

Cybersecurity

Build a Checker using pandas that applies classical machine learning and lightweight LLM calls to vulnerability triage within the Cybersecurity domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA Pinecone · DuckDB

#0040

TRADITIONAL

FOUNDATION

Security Log Analysis Tutorial Notebook

Cybersecurity

Build a Tutorial Notebook using NumPy that applies classical machine learning and lightweight LLM calls to security log analysis within the Cybersecurity domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [MLflow](#) [Gradio](#)

VECTOR / INFRA Weaviate · LanceDB

#0041

TRADITIONAL

FOUNDATION

Incident Response Explorer

Cybersecurity

Build a Explorer using scikit-learn that applies classical machine learning and lightweight LLM calls to incident response within the Cybersecurity domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA Qdrant · OpenSearch

#0042

TRADITIONAL

FOUNDATION

Malware Classification Classifier

Cybersecurity

Build a Classifier using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to malware classification within the Cybersecurity domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	Hugging Face Transformers scikit-learn FastAPI LightGBM
VECTOR / INFRA	Milvus · FAISS

#0043

TRADITIONAL

FOUNDATION

Campaign Copy Generation Predictor

Marketing & Advertising

Build a Predictor using pandas that applies classical machine learning and lightweight LLM calls to campaign copy generation within the Marketing & Advertising domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	pandas Pydantic Polars
VECTOR / INFRA	pgvector · ChromaDB

#0044

TRADITIONAL

FOUNDATION

Audience Segmentation Dashboard

Marketing & Advertising

Build a Dashboard using NumPy that applies classical machine learning and lightweight LLM calls to audience segmentation within the Marketing & Advertising domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	NumPy pandas Docker DuckDB
VECTOR / INFRA	Neo4j · Pinecone

#0045

TRADITIONAL

FOUNDATION

Ad Creative Optimization Basic Chatbot

Marketing & Advertising

Build a Basic Chatbot using scikit-learn that applies classical machine learning and lightweight LLM calls to ad creative optimization within the Marketing & Advertising domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Gemini 2.0 Flash
UI	Gradio
BACKEND	Python · Flask + Gunicorn
AI FRAMEWORKS	scikit-learn Gradio MLflow
VECTOR / INFRA	Elasticsearch · Weaviate

#0046

TRADITIONAL

FOUNDATION

SEO Content Planning Rule Engine

Marketing & Advertising

Build a Rule Engine using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to seo content planning within the Marketing & Advertising domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Mistral 7B
UI	Streamlit + Pandas
BACKEND	Python · FastAPI + Pandas
AI FRAMEWORKS	Hugging Face Transformers scikit-learn XGBoost Streamlit
VECTOR / INFRA	DuckDB · Qdrant

#0047

TRADITIONAL

FOUNDATION

Social Listening Data Pipeline

Marketing & Advertising

Build a Data Pipeline using pandas that applies classical machine learning and lightweight LLM calls to social listening within the Marketing & Advertising domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Phi-4 mini
UI	Jupyter + ipywidgets
BACKEND	Python · Jupyter + FastAPI
AI FRAMEWORKS	pandas LightGBM FastAPI
VECTOR / INFRA	LanceDB · Milvus

#0048

TRADITIONAL

FOUNDATION

Attribution Modeling Visualizer

Marketing & Advertising

Build a Visualizer using NumPy that applies classical machine learning and lightweight LLM calls to attribution modeling within the Marketing & Advertising domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	NumPy pandas Polars Pydantic
VECTOR / INFRA	OpenSearch · pgvector

#0049

TRADITIONAL

FOUNDATION

Ticket Classification Checker

Customer Support

Build a Checker using scikit-learn that applies classical machine learning and lightweight LLM calls to ticket classification within the Customer Support domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	scikit-learn DuckDB Docker
VECTOR / INFRA	FAISS · Neo4j

#0050

TRADITIONAL

FOUNDATION

Virtual Support Agent Tutorial Notebook

Customer Support

Build a Tutorial Notebook using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to virtual support agent within the Customer Support domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	Hugging Face Transformers scikit-learn MLflow Gradio
VECTOR / INFRA	ChromaDB · Elasticsearch

#0051

TRADITIONAL

FOUNDATION

Sentiment Routing Explorer

Customer Support

Build a Explorer using pandas that applies classical machine learning and lightweight LLM calls to sentiment routing within the Customer Support domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA Pinecone · DuckDB

#0052

TRADITIONAL

FOUNDATION

Knowledge Base Search Classifier

Customer Support

Build a Classifier using NumPy that applies classical machine learning and lightweight LLM calls to knowledge base search within the Customer Support domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [FastAPI](#) [LightGBM](#)

VECTOR / INFRA Weaviate · LanceDB

#0053

TRADITIONAL

FOUNDATION

Escalation Prediction Predictor

Customer Support

Build a Predictor using scikit-learn that applies classical machine learning and lightweight LLM calls to escalation prediction within the Customer Support domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA Qdrant · OpenSearch

#0054

TRADITIONAL

FOUNDATION

Voice Support Bot Dashboard

Customer Support

Build a Dashboard using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to voice support bot within the Customer Support domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	Hugging Face Transformers scikit-learn Docker DuckDB
VECTOR / INFRA	Milvus · FAISS

#0055

TRADITIONAL

FOUNDATION

Contract Review Basic Chatbot

Legal & Compliance

Build a Basic Chatbot using pandas that applies classical machine learning and lightweight LLM calls to contract review within the Legal & Compliance domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	pandas Gradio MLflow
VECTOR / INFRA	pgvector · ChromaDB

#0056

TRADITIONAL

FOUNDATION

Legal Research Rule Engine

Legal & Compliance

Build a Rule Engine using NumPy that applies classical machine learning and lightweight LLM calls to legal research within the Legal & Compliance domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	NumPy pandas XGBoost Streamlit
VECTOR / INFRA	Neo4j · Pinecone

#0057

TRADITIONAL

FOUNDATION

Policy Compliance Data Pipeline

Legal & Compliance

Build a Data Pipeline using scikit-learn that applies classical machine learning and lightweight LLM calls to policy compliance within the Legal & Compliance domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [scikit-learn](#) [LightGBM](#) [FastAPI](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0058

TRADITIONAL

FOUNDATION

E-Discovery Visualizer

Legal & Compliance

Build a Visualizer using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to e-discovery within the Legal & Compliance domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [Polars](#) [Pydantic](#)

VECTOR / INFRA DuckDB · Qdrant

#0059

TRADITIONAL

FOUNDATION

Clause Extraction Checker

Legal & Compliance

Build a Checker using pandas that applies classical machine learning and lightweight LLM calls to clause extraction within the Legal & Compliance domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [pandas](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA LanceDB · Milvus

#0060

TRADITIONAL

FOUNDATION

Regulatory Monitoring Tutorial Notebook

Legal & Compliance

Build a Tutorial Notebook using NumPy that applies classical machine learning and lightweight LLM calls to regulatory monitoring within the Legal & Compliance domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS [NumPy](#) [pandas](#) [MLflow](#) [Gradio](#)

VECTOR / INFRA OpenSearch · pgvector

#0061

TRADITIONAL

FOUNDATION

Resume Screening Explorer

Human Resources

Build a Explorer using scikit-learn that applies classical machine learning and lightweight LLM calls to resume screening within the Human Resources domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [scikit-learn](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA FAISS · Neo4j

#0062

TRADITIONAL

FOUNDATION

Interview Assistant Classifier

Human Resources

Build a Classifier using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to interview assistant within the Human Resources domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [FastAPI](#) [LightGBM](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0063

TRADITIONAL

FOUNDATION

Employee Onboarding Predictor

Human Resources

Build a Predictor using pandas that applies classical machine learning and lightweight LLM calls to employee onboarding within the Human Resources domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA Pinecone · DuckDB

#0064

TRADITIONAL

FOUNDATION

Attrition Prediction Dashboard

Human Resources

Build a Dashboard using NumPy that applies classical machine learning and lightweight LLM calls to attrition prediction within the Human Resources domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [Docker](#) [DuckDB](#)

VECTOR / INFRA Weaviate · LanceDB

#0065

TRADITIONAL

FOUNDATION

Skills Gap Analysis Basic Chatbot

Human Resources

Build a Basic Chatbot using scikit-learn that applies classical machine learning and lightweight LLM calls to skills gap analysis within the Human Resources domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [Gradio](#) [MLflow](#)

VECTOR / INFRA Qdrant · OpenSearch

#0066

TRADITIONAL

FOUNDATION

Performance Review Summarizer Rule Engine

Human Resources

Build a Rule Engine using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to performance review summarizer within the Human Resources domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS

Hugging Face Transformers

scikit-learn

XGBoost

Streamlit

VECTOR / INFRA

Milvus · FAISS

#0067

TRADITIONAL

FOUNDATION

Crop Disease Detection Data Pipeline

Agriculture

Build a Data Pipeline using pandas that applies classical machine learning and lightweight LLM calls to crop disease detection within the Agriculture domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS

pandas

LightGBM

FastAPI

VECTOR / INFRA

pgvector · ChromaDB

#0068

TRADITIONAL

FOUNDATION

Yield Prediction Visualizer

Agriculture

Build a Visualizer using NumPy that applies classical machine learning and lightweight LLM calls to yield prediction within the Agriculture domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS

NumPy

pandas

Polars

Pydantic

VECTOR / INFRA

Neo4j · Pinecone

#0069

TRADITIONAL

FOUNDATION

Precision Irrigation Checker

Agriculture

Build a Checker using scikit-learn that applies classical machine learning and lightweight LLM calls to precision irrigation within the Agriculture domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS

scikit-learn

DuckDB

Docker

VECTOR / INFRA

Elasticsearch · Weaviate

#0070

TRADITIONAL

FOUNDATION

Pest Monitoring Tutorial Notebook

Agriculture

Build a Tutorial Notebook using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to pest monitoring within the Agriculture domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS

Hugging Face Transformers

scikit-learn

MLflow

Gradio

VECTOR / INFRA

DuckDB · Qdrant

#0071

TRADITIONAL

FOUNDATION

Soil Health Analytics Explorer

Agriculture

Build a Explorer using pandas that applies classical machine learning and lightweight LLM calls to soil health analytics within the Agriculture domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS

pandas

Streamlit

XGBoost

VECTOR / INFRA

LanceDB · Milvus

#0072

TRADITIONAL

FOUNDATION

Farm Drone Imagery Classifier

Agriculture

Build a Classifier using NumPy that applies classical machine learning and lightweight LLM calls to farm drone imagery within the Agriculture domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	NumPy pandas FastAPI LightGBM
VECTOR / INFRA	OpenSearch · pgvector

#0073

TRADITIONAL

FOUNDATION

Grid Load Forecasting Predictor

Energy & Utilities

Build a Predictor using scikit-learn that applies classical machine learning and lightweight LLM calls to grid load forecasting within the Energy & Utilities domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	scikit-learn Pydantic Polars
VECTOR / INFRA	FAISS · Neo4j

#0074

TRADITIONAL

FOUNDATION

Renewable Output Prediction Dashboard

Energy & Utilities

Build a Dashboard using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to renewable output prediction within the Energy & Utilities domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	Hugging Face Transformers scikit-learn Docker DuckDB
VECTOR / INFRA	ChromaDB · Elasticsearch

#0075

TRADITIONAL

FOUNDATION

Asset Health Monitoring Basic Chatbot

Energy & Utilities

Build a Basic Chatbot using pandas that applies classical machine learning and lightweight LLM calls to asset health monitoring within the Energy & Utilities domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Gemini 2.0 Flash
UI	Gradio
BACKEND	Python · Flask + Gunicorn
AI FRAMEWORKS	pandas Gradio MLflow
VECTOR / INFRA	Pinecone · DuckDB

#0076

TRADITIONAL

FOUNDATION

Energy Trading Rule Engine

Energy & Utilities

Build a Rule Engine using NumPy that applies classical machine learning and lightweight LLM calls to energy trading within the Energy & Utilities domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Mistral 7B
UI	Streamlit + Pandas
BACKEND	Python · FastAPI + Pandas
AI FRAMEWORKS	NumPy pandas XGBoost Streamlit
VECTOR / INFRA	Weaviate · LanceDB

#0077

TRADITIONAL

FOUNDATION

Outage Detection Data Pipeline

Energy & Utilities

Build a Data Pipeline using scikit-learn that applies classical machine learning and lightweight LLM calls to outage detection within the Energy & Utilities domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Phi-4 mini
UI	Jupyter + ipywidgets
BACKEND	Python · Jupyter + FastAPI
AI FRAMEWORKS	scikit-learn LightGBM FastAPI
VECTOR / INFRA	Qdrant · OpenSearch

#0078

TRADITIONAL

FOUNDATION

Smart Meter Analytics Visualizer

Energy & Utilities

Build a Visualizer using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to smart meter analytics within the Energy & Utilities domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	Hugging Face Transformers scikit-learn Polars Pydantic
VECTOR / INFRA	Milvus · FAISS

#0079

TRADITIONAL

FOUNDATION

Citizen Service Assistant Checker

Government & Public Sector

Build a Checker using pandas that applies classical machine learning and lightweight LLM calls to citizen service assistant within the Government & Public Sector domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	pandas DuckDB Docker
VECTOR / INFRA	pgvector · ChromaDB

#0080

TRADITIONAL

FOUNDATION

Document Processing Tutorial Notebook

Government & Public Sector

Build a Tutorial Notebook using NumPy that applies classical machine learning and lightweight LLM calls to document processing within the Government & Public Sector domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	NumPy pandas MLflow Gradio
VECTOR / INFRA	Neo4j · Pinecone

#0081

TRADITIONAL

FOUNDATION

Policy Impact Analysis Explorer

Government & Public Sector

Build a Explorer using scikit-learn that applies classical machine learning and lightweight LLM calls to policy impact analysis within the Government & Public Sector domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [scikit-learn](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0082

TRADITIONAL

FOUNDATION

Emergency Response Classifier

Government & Public Sector

Build a Classifier using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to emergency response within the Government & Public Sector domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [FastAPI](#) [LightGBM](#)

VECTOR / INFRA DuckDB · Qdrant

#0083

TRADITIONAL

FOUNDATION

Permit Review Predictor

Government & Public Sector

Build a Predictor using pandas that applies classical machine learning and lightweight LLM calls to permit review within the Government & Public Sector domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [pandas](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA LanceDB · Milvus

#0084

TRADITIONAL

FOUNDATION

Public Feedback Mining Dashboard

Government & Public Sector

Build a Dashboard using NumPy that applies classical machine learning and lightweight LLM calls to public feedback mining within the Government & Public Sector domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS [NumPy](#) [pandas](#) [Docker](#) [DuckDB](#)

VECTOR / INFRA OpenSearch · pgvector

#0085

TRADITIONAL

FOUNDATION

Content Recommendation Basic Chatbot

Media & Entertainment

Build a Basic Chatbot using scikit-learn that applies classical machine learning and lightweight LLM calls to content recommendation within the Media & Entertainment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [scikit-learn](#) [Gradio](#) [MLflow](#)

VECTOR / INFRA FAISS · Neo4j

#0086

TRADITIONAL

FOUNDATION

Script Generation Rule Engine

Media & Entertainment

Build a Rule Engine using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to script generation within the Media & Entertainment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [XGBoost](#) [Streamlit](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0087

TRADITIONAL

FOUNDATION

Video Editing Assistant Data Pipeline

Media & Entertainment

Build a Data Pipeline using pandas that applies classical machine learning and lightweight LLM calls to video editing assistant within the Media & Entertainment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [LightGBM](#) [FastAPI](#)

VECTOR / INFRA Pinecone · DuckDB

#0088

TRADITIONAL

FOUNDATION

Music Generation Visualizer

Media & Entertainment

Build a Visualizer using NumPy that applies classical machine learning and lightweight LLM calls to music generation within the Media & Entertainment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [Polars](#) [Pydantic](#)

VECTOR / INFRA Weaviate · LanceDB

#0089

TRADITIONAL

FOUNDATION

Subtitle Localization Checker

Media & Entertainment

Build a Checker using scikit-learn that applies classical machine learning and lightweight LLM calls to subtitle localization within the Media & Entertainment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA Qdrant · OpenSearch

#0090

TRADITIONAL

FOUNDATION

Content Moderation Tutorial Notebook

Media & Entertainment

Build a Tutorial Notebook using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to content moderation within the Media & Entertainment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS

Hugging Face Transformers

scikit-learn

MLflow

Gradio

VECTOR / INFRA

Milvus · FAISS

#0091

TRADITIONAL

FOUNDATION

NPC Dialogue Systems Explorer

Gaming

Build a Explorer using pandas that applies classical machine learning and lightweight LLM calls to npc dialogue systems within the Gaming domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS

pandas

Streamlit

XGBoost

VECTOR / INFRA

pgvector · ChromaDB

#0092

TRADITIONAL

FOUNDATION

Procedural Level Generation Classifier

Gaming

Build a Classifier using NumPy that applies classical machine learning and lightweight LLM calls to procedural level generation within the Gaming domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS

NumPy

pandas

FastAPI

LightGBM

VECTOR / INFRA

Neo4j · Pinecone

#0093

TRADITIONAL

FOUNDATION

Player Behavior Analytics Predictor

Gaming

Build a Predictor using scikit-learn that applies classical machine learning and lightweight LLM calls to player behavior analytics within the Gaming domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [scikit-learn](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0094

TRADITIONAL

FOUNDATION

Matchmaking Dashboard

Gaming

Build a Dashboard using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to matchmaking within the Gaming domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [Docker](#) [DuckDB](#)

VECTOR / INFRA DuckDB · Qdrant

#0095

TRADITIONAL

FOUNDATION

Anti-Cheat Detection Basic Chatbot

Gaming

Build a Basic Chatbot using pandas that applies classical machine learning and lightweight LLM calls to anti-cheat detection within the Gaming domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [pandas](#) [Gradio](#) [MLflow](#)

VECTOR / INFRA LanceDB · Milvus

#0096

TRADITIONAL

FOUNDATION

Game Testing Bots Rule Engine

Gaming

Build a Rule Engine using NumPy that applies classical machine learning and lightweight LLM calls to game testing bots within the Gaming domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS NumPy pandas XGBoost Streamlit

VECTOR / INFRA OpenSearch · pgvector

#0097

TRADITIONAL

FOUNDATION

Pick-and-Place Control Data Pipeline

Robotics

Build a Data Pipeline using scikit-learn that applies classical machine learning and lightweight LLM calls to pick-and-place control within the Robotics domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS scikit-learn LightGBM FastAPI

VECTOR / INFRA FAISS · Neo4j

#0098

TRADITIONAL

FOUNDATION

SLAM Navigation Visualizer

Robotics

Build a Visualizer using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to slam navigation within the Robotics domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS Hugging Face Transformers scikit-learn Polars Pydantic

VECTOR / INFRA ChromaDB · Elasticsearch

#0099

TRADITIONAL

FOUNDATION

Manipulation Planning Checker

Robotics

Build a Checker using pandas that applies classical machine learning and lightweight LLM calls to manipulation planning within the Robotics domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA Pinecone · DuckDB

#0100

TRADITIONAL

FOUNDATION

Human-Robot Collaboration Tutorial Notebook

Robotics

Build a Tutorial Notebook using NumPy that applies classical machine learning and lightweight LLM calls to human-robot collaboration within the Robotics domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [MLflow](#) [Gradio](#)

VECTOR / INFRA Weaviate · LanceDB

#0101

TRADITIONAL

FOUNDATION

Swarm Coordination Explorer

Robotics

Build a Explorer using scikit-learn that applies classical machine learning and lightweight LLM calls to swarm coordination within the Robotics domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA Qdrant · OpenSearch

#0102

TRADITIONAL

FOUNDATION

Sim-to-Real Transfer Classifier

Robotics

Build a Classifier using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to sim-to-real transfer within the Robotics domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS

Hugging Face Transformers

scikit-learn

FastAPI

LightGBM

VECTOR / INFRA

Milvus · FAISS

#0103

TRADITIONAL

FOUNDATION

Perception Fusion Predictor

Autonomous Vehicles

Build a Predictor using pandas that applies classical machine learning and lightweight LLM calls to perception fusion within the Autonomous Vehicles domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS

pandas

Pydantic

Polars

VECTOR / INFRA

pgvector · ChromaDB

#0104

TRADITIONAL

FOUNDATION

Trajectory Prediction Dashboard

Autonomous Vehicles

Build a Dashboard using NumPy that applies classical machine learning and lightweight LLM calls to trajectory prediction within the Autonomous Vehicles domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS

NumPy

pandas

Docker

DuckDB

VECTOR / INFRA

Neo4j · Pinecone

#0105

TRADITIONAL

FOUNDATION

Driver Monitoring Basic Chatbot

Autonomous Vehicles

Build a Basic Chatbot using scikit-learn that applies classical machine learning and lightweight LLM calls to driver monitoring within the Autonomous Vehicles domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Gemini 2.0 Flash
UI	Gradio
BACKEND	Python · Flask + Gunicorn
AI FRAMEWORKS	scikit-learn Gradio MLflow
VECTOR / INFRA	Elasticsearch · Weaviate

#0106

TRADITIONAL

FOUNDATION

HD Map Generation Rule Engine

Autonomous Vehicles

Build a Rule Engine using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to hd map generation within the Autonomous Vehicles domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Mistral 7B
UI	Streamlit + Pandas
BACKEND	Python · FastAPI + Pandas
AI FRAMEWORKS	Hugging Face Transformers scikit-learn XGBoost Streamlit
VECTOR / INFRA	DuckDB · Qdrant

#0107

TRADITIONAL

FOUNDATION

Scenario Simulation Data Pipeline

Autonomous Vehicles

Build a Data Pipeline using pandas that applies classical machine learning and lightweight LLM calls to scenario simulation within the Autonomous Vehicles domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Phi-4 mini
UI	Jupyter + ipywidgets
BACKEND	Python · Jupyter + FastAPI
AI FRAMEWORKS	pandas LightGBM FastAPI
VECTOR / INFRA	LanceDB · Milvus

#0108

TRADITIONAL

FOUNDATION

Fleet Teleoperation Visualizer

Autonomous Vehicles

Build a Visualizer using NumPy that applies classical machine learning and lightweight LLM calls to fleet teleoperation within the Autonomous Vehicles domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Claude 3.5 Haiku

UI Streamlit Multi-page

BACKEND Python · FastAPI (single-file)

AI FRAMEWORKS [NumPy](#) [pandas](#) [Polars](#) [Pydantic](#)

VECTOR / INFRA OpenSearch · pgvector

#0109

TRADITIONAL

FOUNDATION

Literature Synthesis Checker

Scientific Research

Build a Checker using scikit-learn that applies classical machine learning and lightweight LLM calls to literature synthesis within the Scientific Research domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM GPT-4o mini

UI Streamlit

BACKEND Python · FastAPI + Uvicorn

AI FRAMEWORKS [scikit-learn](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA FAISS · Neo4j

#0110

TRADITIONAL

FOUNDATION

Protein Structure Prediction Tutorial Notebook

Scientific Research

Build a Tutorial Notebook using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to protein structure prediction within the Scientific Research domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Llama 3.2 3B (local)

UI Streamlit + Plotly

BACKEND Python · FastAPI + SQLite

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [MLflow](#) [Gradio](#)

VECTOR / INFRA ChromaDB · Elasticsearch

#0111

TRADITIONAL

FOUNDATION

Experiment Design Explorer

Scientific Research

Build a Explorer using pandas that applies classical machine learning and lightweight LLM calls to experiment design within the Scientific Research domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [pandas](#) [Streamlit](#) [XGBoost](#)

VECTOR / INFRA Pinecone · DuckDB

#0112

TRADITIONAL

FOUNDATION

Lab Automation Classifier

Scientific Research

Build a Classifier using NumPy that applies classical machine learning and lightweight LLM calls to lab automation within the Scientific Research domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [NumPy](#) [pandas](#) [FastAPI](#) [LightGBM](#)

VECTOR / INFRA Weaviate · LanceDB

#0113

TRADITIONAL

FOUNDATION

Materials Discovery Predictor

Scientific Research

Build a Predictor using scikit-learn that applies classical machine learning and lightweight LLM calls to materials discovery within the Scientific Research domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [scikit-learn](#) [Pydantic](#) [Polars](#)

VECTOR / INFRA Qdrant · OpenSearch

#0114

TRADITIONAL

FOUNDATION

Research Data Analysis Dashboard

Scientific Research

Build a Dashboard using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to research data analysis within the Scientific Research domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	Hugging Face Transformers scikit-learn Docker DuckDB
VECTOR / INFRA	Milvus · FAISS

#0115

TRADITIONAL

FOUNDATION

Emissions Monitoring Basic Chatbot

Climate & Environment

Build a Basic Chatbot using pandas that applies classical machine learning and lightweight LLM calls to emissions monitoring within the Climate & Environment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	GPT-4o mini
UI	Streamlit
BACKEND	Python · FastAPI + Uvicorn
AI FRAMEWORKS	pandas Gradio MLflow
VECTOR / INFRA	pgvector · ChromaDB

#0116

TRADITIONAL

FOUNDATION

Wildfire Prediction Rule Engine

Climate & Environment

Build a Rule Engine using NumPy that applies classical machine learning and lightweight LLM calls to wildfire prediction within the Climate & Environment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Llama 3.2 3B (local)
UI	Streamlit + Plotly
BACKEND	Python · FastAPI + SQLite
AI FRAMEWORKS	NumPy pandas XGBoost Streamlit
VECTOR / INFRA	Neo4j · Pinecone

#0117

TRADITIONAL

FOUNDATION

Biodiversity Tracking Data Pipeline

Climate & Environment

Build a Data Pipeline using scikit-learn that applies classical machine learning and lightweight LLM calls to biodiversity tracking within the Climate & Environment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Gemini 2.0 Flash

UI Gradio

BACKEND Python · Flask + Gunicorn

AI FRAMEWORKS [scikit-learn](#) [LightGBM](#) [FastAPI](#)

VECTOR / INFRA Elasticsearch · Weaviate

#0118

TRADITIONAL

FOUNDATION

Carbon Accounting Visualizer

Climate & Environment

Build a Visualizer using Hugging Face Transformers that applies classical machine learning and lightweight LLM calls to carbon accounting within the Climate & Environment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Mistral 7B

UI Streamlit + Pandas

BACKEND Python · FastAPI + Pandas

AI FRAMEWORKS [Hugging Face Transformers](#) [scikit-learn](#) [Polars](#) [Pydantic](#)

VECTOR / INFRA DuckDB · Qdrant

#0119

TRADITIONAL

FOUNDATION

Weather Forecasting Checker

Climate & Environment

Build a Checker using pandas that applies classical machine learning and lightweight LLM calls to weather forecasting within the Climate & Environment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM Phi-4 mini

UI Jupyter + ipywidgets

BACKEND Python · Jupyter + FastAPI

AI FRAMEWORKS [pandas](#) [DuckDB](#) [Docker](#)

VECTOR / INFRA LanceDB · Milvus

Ocean Data Analytics Tutorial Notebook

Climate & Environment

Build a Tutorial Notebook using NumPy that applies classical machine learning and lightweight LLM calls to ocean data analytics within the Climate & Environment domain. Focus on data cleaning, feature engineering, baseline metrics and a clear evaluation notebook.

LLM	Claude 3.5 Haiku
UI	Streamlit Multi-page
BACKEND	Python · FastAPI (single-file)
AI FRAMEWORKS	NumPy pandas MLflow Gradio
VECTOR / INFRA	OpenSearch · pgvector

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